

Dream-to-Center: A Research Journey Toward Predictive One-Axis Gimbal Servoing

Autonomous Observation Lab — Technical Research Retrospective

Project period covered: 26–31 August 2026

Status: synthetic research prototype; not a deployment or novelty claim

Abstract

This report traces the development of a configurable predictive visual-servo system for a custom one-axis camera gimbal intended for quadcopter deployment. The task is to keep a designated bounding box centered despite target motion, vehicle rotation, detector delay and dropout, and uncertain actuator dynamics. The implemented system combines a 36,240-parameter causal gated recurrent unit (GRU), trained with privileged simulator target-state labels but restricted to deployable inputs at runtime, with conventional desired-rate or absolute-position command adapters and an independent bounded servo model.

Across three independent training initializations and 24 paired randomized test worlds, the disturbance-aware GRU improves all declared core tracking metrics over an analytical constant-velocity controller in both command modes. For rate control, mean error falls from 22.84° to $18.57 \pm 0.27^\circ$, episode P95 from 47.10° to $38.12 \pm 0.58^\circ$, and loss of view from 27.06% to $19.84 \pm 0.43\%$. Position control shows smaller but replicated gains. These improvements are strongest under high latency and aggressive motion and vanish at a physically unreachable travel-limit ceiling.

The result is promising but incomplete. Rate commands are less smooth, post-loss recovery is slower in the original closed-loop comparison, and every directed recovery design tested so far fails a fresh safety gate. Uncertainty scaling improves held-out 2σ coverage but does not fully calibrate the residual distribution. The evidence therefore supports a narrow conclusion: learned temporal target-state inference outperforms the present reactive and model-based baselines in randomized simulation, but recovery, absolute mission requirements, embedded timing, recorded flight motion, and physical hardware transfer remain unresolved.

Keywords: visual servoing, gimbal control, recurrent neural network, privileged supervision, disturbance awareness, sim-to-real, uncertainty, loss-of-view recovery

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1. Introduction

A camera mounted on a moving quadcopter does not observe target motion in isolation. Its image-plane error combines at least four causes:

1. target motion in the world;
2. vehicle body rotation;
3. delayed and imperfect gimbal response; and
4. delayed, noisy, or missing detector measurements.

A proportional controller can react to the current bounding-box displacement, but it cannot determine which hidden cause produced that displacement. A constant-velocity estimator adds explicit temporal structure, yet it assumes that recent motion will continue. That assumption becomes fragile during body maneuvers, reversals, prolonged dropouts, and uncertain actuator lag.

The project began with the question:

Can a compact recurrent visual servo infer enough hidden target, disturbance, delay, and actuator state from causal deployment signals to reduce tail error and loss of view under unseen motion and hardware variation?

The initial concept proposed a complete deterministic recurrent actor, potentially fine-tuned with continuous-control reinforcement learning. The work completed so far deliberately stops earlier. The validated learned component is a supervised causal GRU target-state estimator. Conventional, configurable adapters convert its estimated bearing and angular rate into rate or position commands. This separation lets us test whether learned temporal state is useful before attributing gains to end-to-end policy optimization.

The contribution at this stage is therefore an experimental package rather than a finished controller claim:

- a deterministic configurable gimbal benchmark and visualizer;
- a common target-state interface for analytical, learned, and privileged estimators;
- a leakage-controlled privileged dataset pipeline;
- a compact causal GRU with uncertainty outputs;
- paired rate/position closed-loop comparisons and multi-seed replication;
- uncertainty and recovery safety protocols that preserve negative results; and
- exact replay and mouse-openable verification dashboards.

Target centering, predictive pan-tilt control, and learned visual servoing are established research areas [1–4]. The project does not claim that using AI on a gimbal is novel. Its candidate research value lies in controlled evidence about when deployable recurrent state helps under disturbance, delay, and plant variation.

2. Problem formulation and system boundary

The study controls one rotational axis. Zero gimbal position always means body forward. All camera and actuator quantities—including field of view, frame rate, detector latency, travel, rate, acceleration, lag, deadband, quantization, and command delay—are configuration values rather than embedded hardware assumptions.

For a selected normalized image coordinate $e_t \in [-1, 1]$, the deployable observation is drawn from

$$o_t = (e_t, w_t, h_t, c_t, m_t, \Delta t_t, q_t, \dot{q}_t, \omega_t^{\text{body}}, u_{t-1})$$

with an explicit validity mask for every optional signal. The estimator emits

$$\hat{s}_t = (\hat{\theta}_t^{\text{target/body}}, \dot{\hat{\theta}}_t^{\text{target/body}}, \sigma_{\theta,t}, \sigma_{\dot{\theta},t}, m_t)$$

including measurement time and prediction horizon. A configured adapter then produces one bounded logical action:

$$u_t^{\text{rate}} = \text{clip}\left(\frac{\dot{\hat{\theta}}_t + k_p(\hat{\theta}_t - q_t)}{\dot{q}_{\max}}, -1, 1\right)$$

or an absolute body-relative position command obtained by mapping the bounded target bearing into the configured asymmetric travel interval. The simulated or embedded inner loop remains responsible for realizing that command safely.

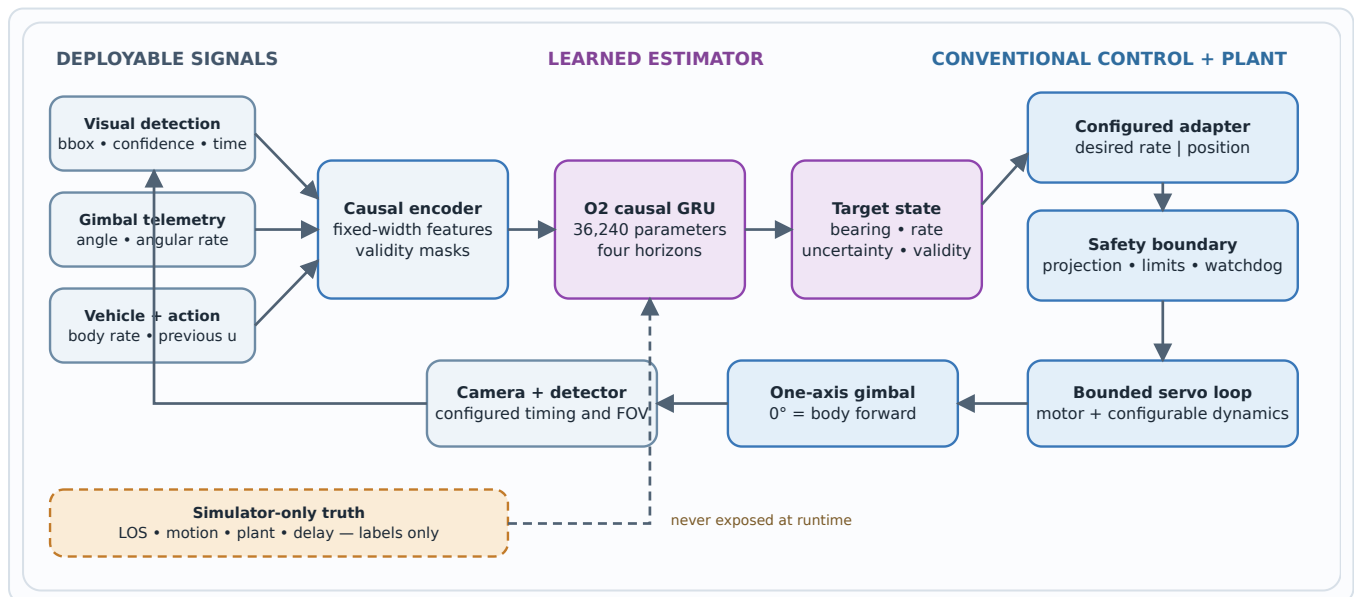


Figure 1. System boundary. Privileged truth supervises learning but is never an actor input. The GRU estimates target state; conventional adapters and the inner servo retain the command and safety boundary.

Three observation profiles isolate the value of deployment telemetry:

Profile	Runtime inputs	Experimental role
O0 vision-only	bbox, timing, confidence, previous action	hardest partial-observation ablation
O1 servo-aware	O0 plus gimbal angle and rate	payload-local deployment profile
O2 disturbance-aware	O1 plus vehicle body rate	strongest currently tested profile
OP privileged	true LOS, motion, plant and delay state	labels and ceiling only

3. The research journey

The original five-step program has been completed and extended through replication, uncertainty, and recovery safety testing.

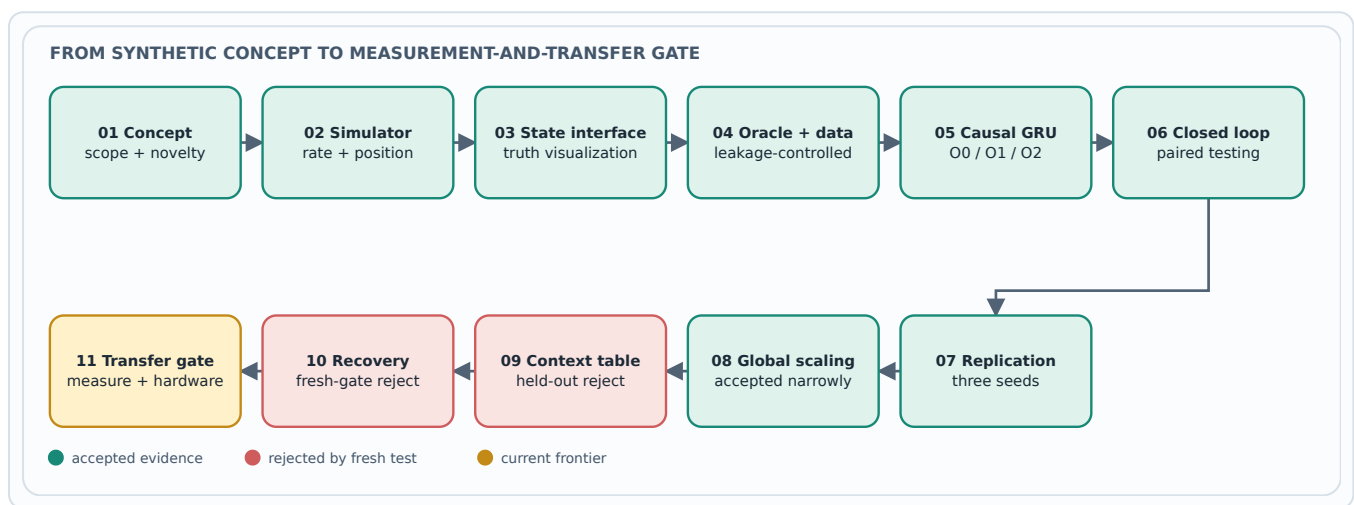


Figure 2. Development trajectory. Rejected experiments were retained rather than silently optimized away. The current transition is from synthetic policy validation to measurement and transfer.

Planned step	Implemented result	Status
Extract analytical predictor into a common state interface	timestamped bearing/rate/uncertainty contract plus truth visualization	complete
Build a latency, dropout, lag, motion, saturation and recovery suite	six closed-loop families plus seven expanded recovery families	complete
Add a privileged oracle	separate labels and rate/position ceiling controller	complete
Train a compact causal temporal model	64-state, one-layer GRU at four horizons	complete
Compare observation profiles and baselines	O0/O1/O2 open-loop, paired closed-loop, rate/position and three seeds	complete
Calibrate uncertainty	accepted global scaling; rejected contextual table	complete
Establish safe recovery	multiple mechanisms evaluated; none deployment-eligible	open
Transfer to the custom gimbal	telemetry and hardware parameters not yet recorded	next

4. Experimental method

4.1 Configurable plant and sensing

Each episode combines independently randomized target and vehicle angular motion with a one-axis gimbal plant and camera/detector process. The plant supports rate and acceleration limits, command latency, first-order rate lag, deadband, quantization, and hard travel limits. The camera supports configurable field of view, sample rate, detection latency and jitter, center/size noise, miss probability, and scheduled dropout intervals.

The primary closed-loop suite contains six families: nominal combined motion, high latency, detector dropout/noise, slow servo, aggressive motion, and travel-limit recovery. The later recovery suite adds detector micro-bursts, target reversal during outage, negative-side re-entry, and body-maneuver outages. Identical target motion, body motion, detector randomness, and hardware realization are paired across controllers.

4.2 Privileged data without runtime leakage

The oracle receives simulator motion and diagnostics and produces body-relative target bearing and angular rate at 0.0, 0.1, 0.2, and 0.3 s horizons. It also provides ideal rate and position targets through the same simulated actuator limits. Oracle state is stored separately from deployable features and cannot enter the runtime estimator.

The mixed-command dataset contains 432 training, 144 validation, and 144 test episodes, representing 72/24/24 distinct randomized world-and-plant variants. Collection uses proportional, analytical-predictive, and privileged-oracle behaviors in both rate and position modes. Split seed blocks and configuration hashes are disjoint and recorded in manifests.

4.3 Learned estimator

The implemented model has a 64-dimensional feature embedding, a one-layer unidirectional 64-state GRU, and Gaussian bearing/rate heads at four horizons, for 36,240 trainable parameters. All models train for 50 epochs with best-validation checkpoint restoration. A validation controller selects the forecast horizon separately for each observation profile and command mode before the test split is accessed.

The learned model is causal and action-conditioned. It never receives target truth, future motion, servo queue state, simulator episode identity, or absolute simulation time as a feature.

4.4 Baselines and metrics

The implemented controller comparison contains:

- delayed proportional bbox feedback;
- an analytical constant-velocity target-state estimator;
- servo-aware O1 and disturbance-aware O2 GRUs; and
- a privileged oracle as a dataset/control ceiling, not a deployable baseline.

The analytical baseline is stronger than a bare heuristic: it uses calibrated camera geometry, measurement history, uncertainty, and rate feed-forward. The primary research comparison is therefore O2 versus analytical control, with proportional control retained as a secondary reference.

Reported metrics include mean and P95 tracking error, loss-of-view fraction, recovery time and events, saturation, command magnitude, command variation, and a configured control cost combining normalized error, visibility loss, effort, and action change.

5. Results

5.1 Observation-profile and prediction evidence

On the complete randomized prediction test set, O1 gives the lowest bearing RMSE while O2 gives the lowest angular-rate RMSE:

Profile	Bearing RMSE	Rate RMSE	Bearing 2σ	Rate 2σ
O0 vision-only	22.66°	35.20°/s	96.69%	92.46%
O1 servo-aware	21.47°	35.22°/s	93.04%	92.11%
O2 disturbance-aware	22.01°	31.21°/s	94.75%	92.82%

On exactly the subset where the analytical estimator is valid, O2 improves bearing/rate RMSE from 11.12°/47.11°/s to 8.20°/32.59°/s. Explicit geometry is slightly better at immediate bearing, while O2 becomes increasingly stronger at 0.1–0.3 s horizons. The learned model is available during more difficult intervals, but its error grows sharply when no current observation identifies an off-screen target. Recurrence improves inference; it does not create missing information.

5.2 Closed-loop tracking and replication

The strongest current evidence is a three-initialization replication. Training seeds 17, 29, and 43 use the same frozen architecture, optimizer, data, horizon selection protocol, and 24 paired test worlds. No best seed is selected.

Closed-loop tracking: analytical baseline vs replicated O2 GRU

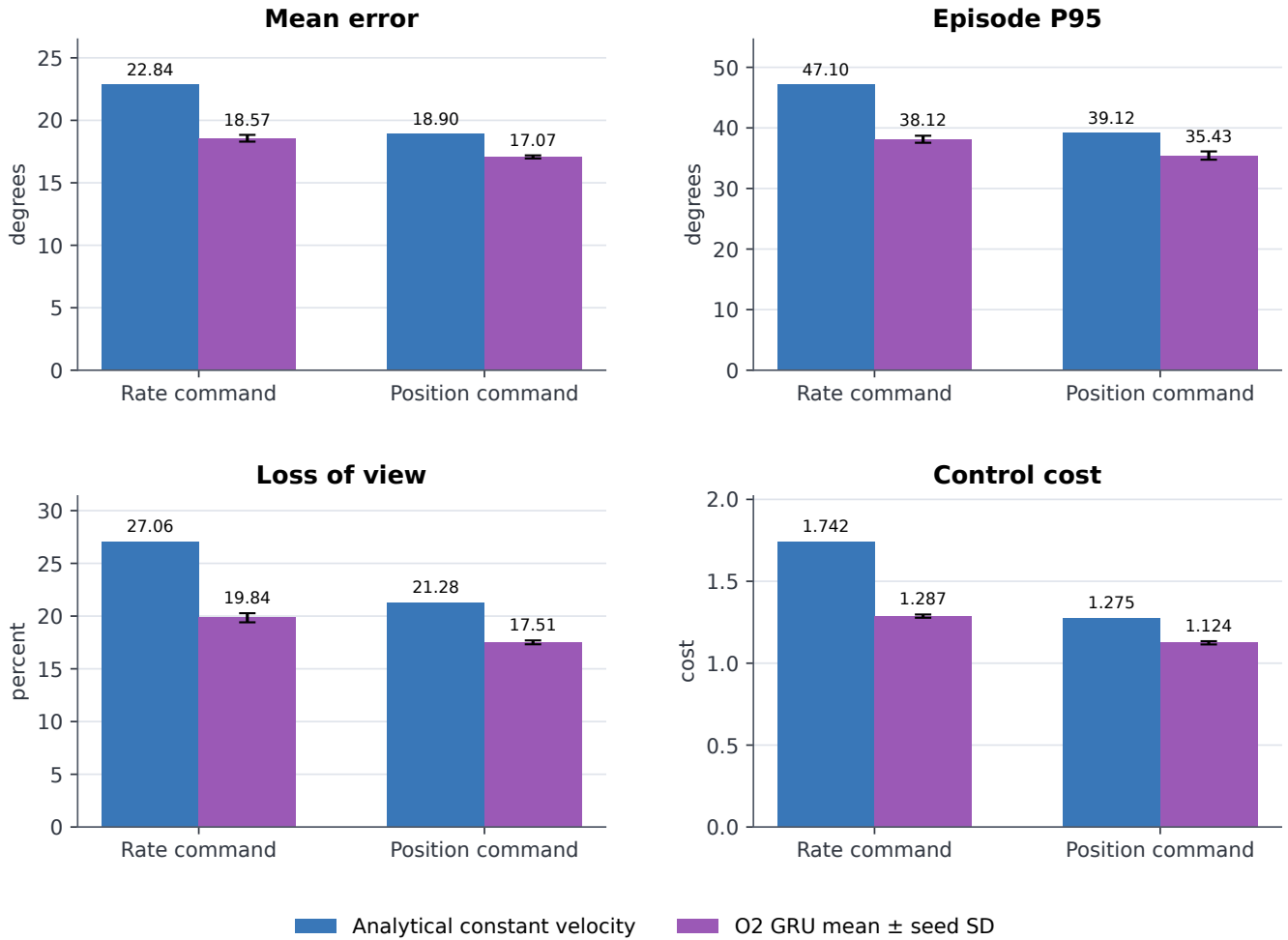


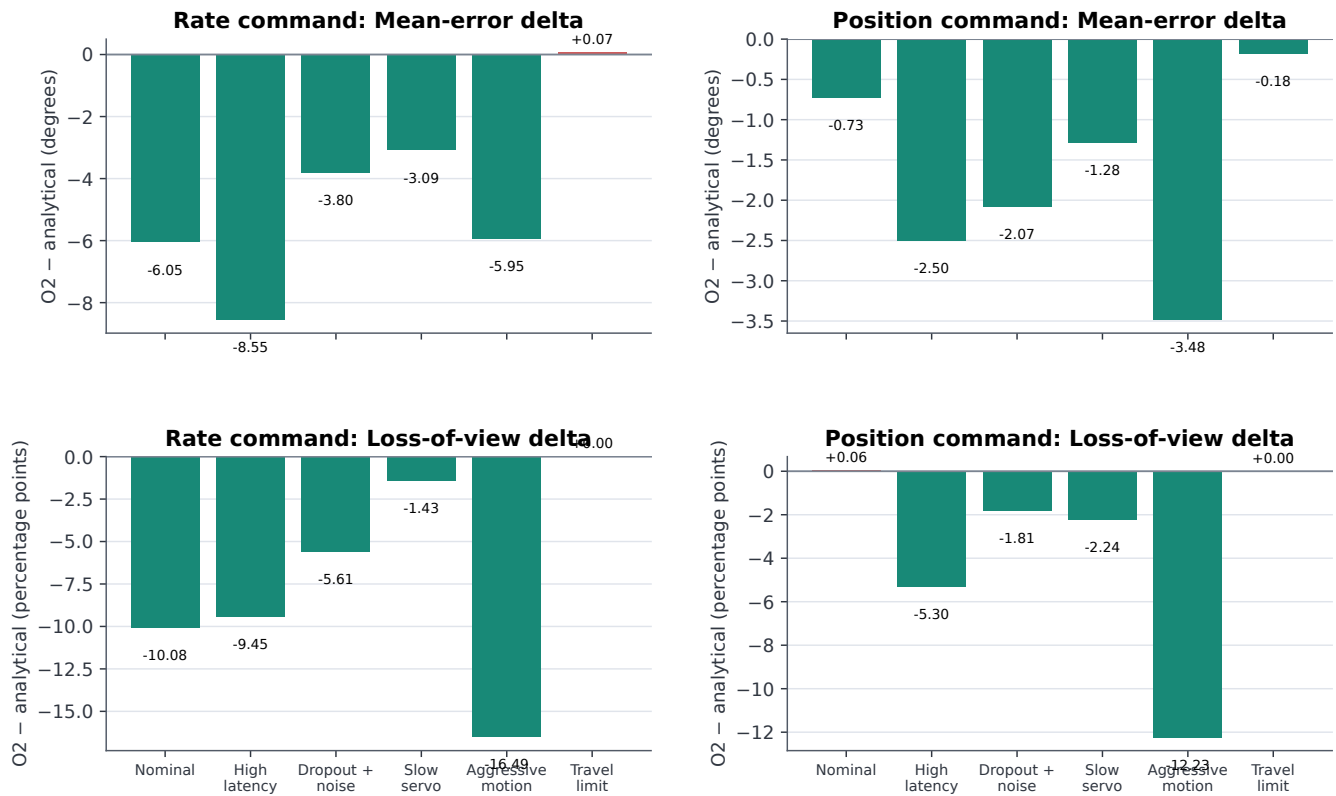
Figure 3. Replicated closed-loop performance. Error bars show sample standard deviation across independent GRU initializations. Every seed improves all four core metrics in both command modes.

Mode / metric	Analytical	O2 mean $\pm$ seed SD	Mean change
Rate mean error	22.84°	18.57 $\pm$ 0.27°	-4.27°
Rate episode P95	47.10°	38.12 $\pm$ 0.58°	-8.98°
Rate loss of view	27.06%	19.84 $\pm$ 0.43%	-7.22 pp
Rate control cost	1.742	1.287 $\pm$ 0.010	-0.455
Position mean error	18.90°	17.07 $\pm$ 0.12°	-1.83°
Position episode P95	39.12°	35.43 $\pm$ 0.68°	-3.69°
Position loss of view	21.28%	17.51 $\pm$ 0.18%	-3.77 pp
Position control cost	1.275	1.124 $\pm$ 0.010	-0.151

All rate experiments select the current-state GRU output. Their advantage comes from better recurrent bearing/rate state and angular-rate feed-forward rather than leading the loop with a future bearing target. Position experiments select 0.2 s for seed 17 and 0.1 s for seeds 29 and 43. Position performance replicates, but a single predictive horizon does not.

5.3 Where the advantage occurs

Where the learned controller helps—and where geometry dominates



Green bars favor O2; red bars indicate regression. Frozen seed-17 controller artifact.

Figure 4. Scenario decomposition. The learned estimator provides its largest gains under high latency and aggressive motion. Both methods converge at the travel-limit case because the target leaves the reachable mechanical and camera envelope.

On the frozen seed-17 controller artifact, rate O2 reduces mean error by 8.55° in high latency and 5.95° under aggressive motion. Loss of view falls by 9.45 and 16.49 percentage points, respectively. Position control shows the same pattern at smaller magnitude. Rate control has a negligible $+0.07^\circ$ regression in the travel-limit family; position loss of view has a negligible +0.06-point nominal regression. Paired episode plots in the performance dashboard expose additional isolated regressions hidden by scenario averages.

5.4 Uncertainty calibration

The raw O2 model is slightly overconfident. A positive scale is fitted for each horizon and output dimension using validation residuals, then frozen before test. Scales range from 1.007 to 1.073.

Global uncertainty scaling improves held-out tail coverage

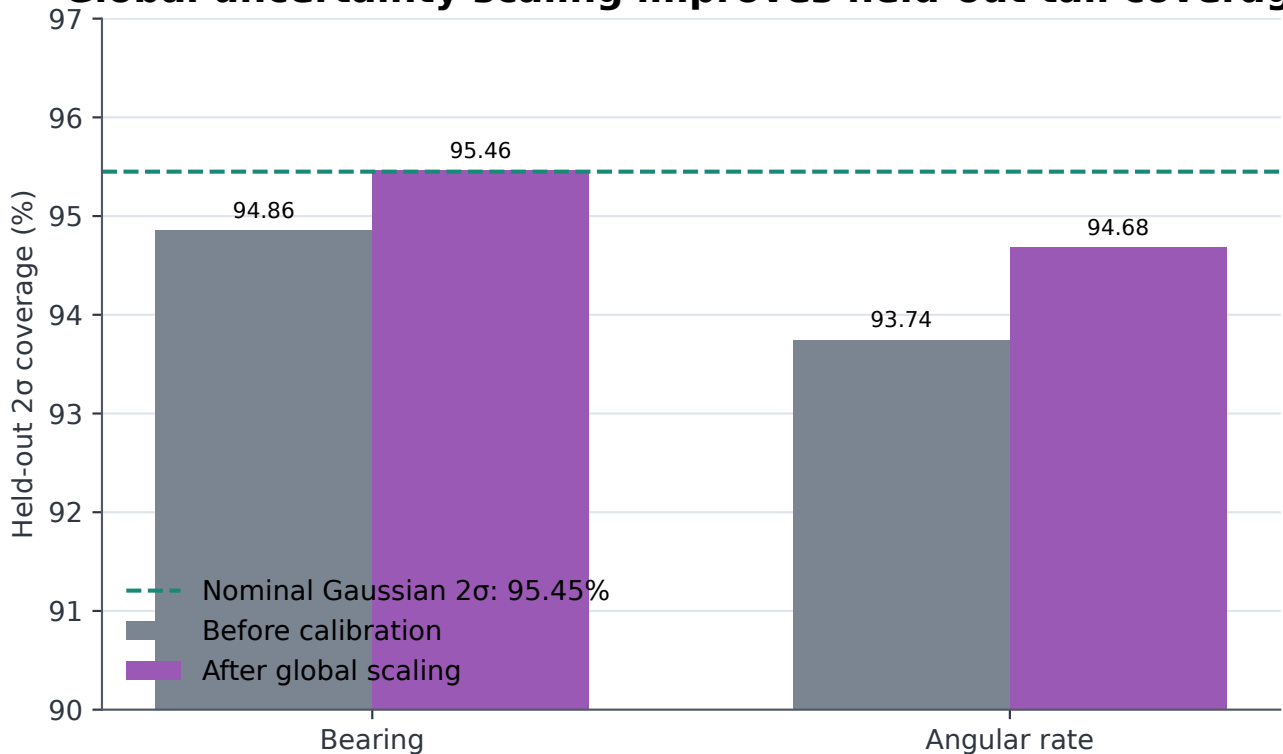


Figure 5. Held-out 2σ coverage. Bearing becomes essentially nominal; rate improves but remains modestly under-covered.

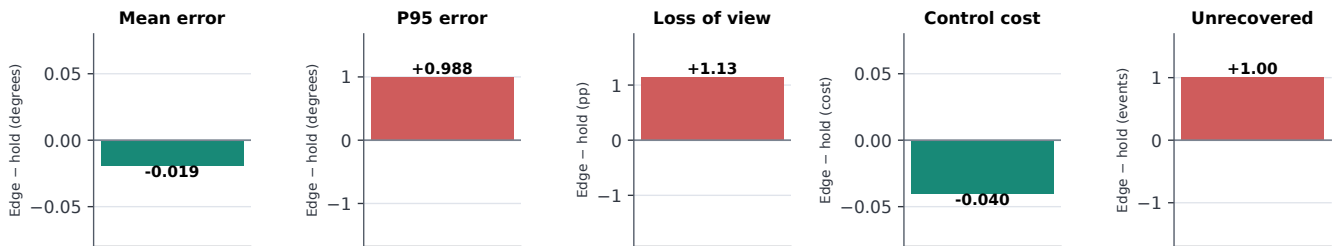
Calibration improves test bearing coverage from 94.86% to 95.46% and rate coverage from 93.74% to 94.68%, while also improving Gaussian negative log-likelihood. This is not complete distribution calibration: mean absolute calibration error across several central intervals becomes worse, and residuals remain regime-dependent. A more complex contextual calibrator improved validation likelihood but failed held-out generalization and was rejected.

5.5 Loss-of-view recovery

Recovery produced the clearest negative results. A blind sweep slightly changed aggregate visibility while worsening error, cost, and terminal failures. A belief-guided state machine improved some average metrics in an initial small suite but introduced recoverable detector-burst failures. The expanded suite then exposed a structural flaw: stale constant-rate projection continued in the wrong direction during target and body reversals. Native hold produced 16 development failures; ungated belief search produced 24.

An edge-conditioned design permitted search only when the last valid bbox was near the image edge and moving outward. It repaired all eight excess development failures and reduced control cost. The configuration was frozen before evaluation on new 46000-series seeds.

Fresh recovery safety gate: REJECT



Lower is better. Cost and mean error improve slightly, but tail error, visibility, and events regress.

Figure 6. Recovery safety result. Edge conditioning improves mean error and cost slightly, but worsens P95, visibility, and unrecovered-event count. The fresh deployment gate rejects it.

The extra failure occurs in a detector-burst variant where edge motion falsely resembles a physical exit. Retuning on the observed fresh block would be test leakage, so native hold remains the accepted fallback.

6. Discussion

6.1 Is the learned controller better than a heuristic?

Yes, within the tested synthetic distribution—and the evidence is stronger than a comparison with proportional feedback alone. O2 beats both delayed proportional control and the analytical constant-velocity target-state controller on mean error, tail error, visibility, and declared cost. The gains survive three training initializations and occur where temporal inference is expected to matter.

The precise statement is:

A compact disturbance-aware recurrent target-state estimator, coupled to the same conventional command adapters, improves randomized synthetic closed-loop tracking over the present reactive and analytical baselines.

It would be inaccurate to call the current system a fully learned end-to-end policy or a completed RL controller. The learned component estimates state; the control law remains explicit and configurable.

6.2 Smoothness and actuator burden

Rate-command variation is consistently worse on average: analytical control is 2.301 changes/s versus 2.468 ± 0.109 for O2, a $+0.167/s$ increase. Position variation averages 1.012 ± 0.157 versus 1.032 analytically, but the finding is not stable: seed 17 is $+0.155/s$ worse while other seeds compensate. Saturation also does not improve consistently.

Thus the learned system is not simply better on every axis. Its current benefit is tracking and visibility, not guaranteed smoothness. A future controller must either include an explicit jerk/bandwidth requirement, train with a matched smoothness objective, or filter commands without erasing the temporal gain.

6.3 Prevention is not recovery

In the original seed-17 comparison, O2 reduces how often the target is lost but takes longer to recover once loss occurs: 1.58 s versus 1.07 s for rate and 1.38 s versus 0.98 s for position. This separates two research problems:

- recurrent estimation improves loss prevention while observations remain informative;
- reacquisition after observability is lost requires a different belief, search, or sensing mechanism.

The recovery experiments show why aggregate averages cannot replace safety gates. Blind or directed search may reduce failures in one scenario family while creating failures in another.

6.4 Absolute performance remains undefined

The project has a relative research target—beat paired baselines—but not yet an absolute mission specification. Mean errors of 17–19° and loss-of-view rates of 18–20% may be acceptable for a deliberately difficult synthetic benchmark and still be unacceptable for a real observation mission. Requirements must be stated before tuning the next controller:

Deployment requirement	Current target
Maximum nominal mean centering error	TBD
Maximum nominal and shifted P95 error	TBD
Maximum loss-of-view fraction and event rate	TBD
Maximum recovery time and unrecovered events	TBD
Maximum command variation, rate and acceleration	TBD
Maximum inference mean/P99 latency	TBD
Allowed camera/servo calibration error	TBD
Required fallback behavior	native hold provisionally

7. Acceptance ledger

Claim or gate	Evidence	Verdict
Configurable hardware-independent simulation	camera, detector, timing, servo and both adapters serialized	pass
Learned state beats proportional control	paired randomized test	pass
Learned state beats analytical constant velocity	four core metrics, both adapters	pass
Result survives model initialization	three independent O2 seeds	pass
Advantage appears under temporal stress	high-latency and aggressive-motion decomposition	pass
Stable future-bearing mechanism	rate chooses 0 s; position horizon varies	not established
Smoothness/saturation advantage	rate variation worse; mixed position/saturation	fail/partial
Reliable uncertainty	better NLL and 2 σ ; worse multi-level MACE	partial
Safe directed recovery	repeated fresh-test regressions	fail
Strong final baseline set	robust PID/MPC and recurrent model-free RL absent	incomplete
Embedded inference budget	no target compute measurement	untested
Recorded-flight and real-gimbal transfer	no platform data or identified hardware yet	untested

8. Threats to validity

1. **Synthetic motion.** Six authored families and randomized parameters do not reproduce the full spectrum, coupling, vibration, and nonstationarity of flight data.
2. **Small replication count.** Three model initializations establish a useful consistency check, not a population-level training analysis.
3. **Frozen but limited test set.** The primary test contains 24 paired world/plant variants. Later recovery protocols use larger, disjoint blocks but remain synthetic.
4. **Baseline incompleteness.** The analytical estimator is credible, but the planned tuned PID, identified MPC, recurrent SAC/TD3, and equal-budget learned baselines are not complete.
5. **One-dimensional geometry.** Axis coupling, rolling-shutter effects, two-axis kinematics, and vehicle translation are excluded.
6. **Feature-level perception.** The experiments begin with bbox measurements; detector identity switches, systematic bias, and real confidence behavior are not represented.
7. **No embedded timing.** Model size is known, but mean and tail inference latency, memory, scheduling jitter, and communication delay are not measured on target compute.
8. **No hardware safety evidence.** Constraint logic is tested in simulation, not against the electrical, mechanical, or firmware behavior of the custom servo.

9. Recommended next phase

The next milestone should be a measurement-and-transfer gate rather than more synthetic recovery tuning.

1. Define a versioned deployment telemetry contract covering timestamps, bbox, detection validity, gimbal feedback, typed command, and optional body rate.
2. Define the absolute requirement table above with mission owners.
3. Identify the camera and servo: FOV, frame timing, latency, travel, maximum rate/acceleration, lag, deadband, quantization, and mounting sign.
4. Implement recorded-trace ingestion and exact open-loop replay for body, gimbal, detector, and bbox logs.
5. Replay recorded flight motion through the current simulator and quantify distribution mismatch before retraining.
6. Measure GRU mean/P95/P99 inference latency and memory on the intended compute platform.
7. Close the baseline set with tuned PID/observer/MPC and a matched recurrent model-free learner.
8. Proceed through hardware-in-the-loop, oscillating bench, restrained flight, and free flight only with explicit safety gates and native hold available.

10. Reproducibility and inspection

Generate the exact-data figures with:

```
python3 scripts/generate_gimbal_paper_figures.py
```

Install the optional reporting dependency when needed:

```
python3 -m pip install -e '.[reporting]'
```

Build the print-ready PDF locally with Chrome or Chromium:

```
python3 scripts/build_gimbal_journey_pdf.py
```

Open the consolidated comparison without typing commands by launching `scripts/open_gimbal_performance_dashboard.sh`. The dashboard contains the aggregate tables, per-scenario curves, every paired test delta, and training seed stability. Recovery and uncertainty have separate exact-replay dashboards.

Primary frozen artifacts:

- `artifacts/gimbal_mixed_gru_closed_loop_comparison.json`;
- `artifacts/gimbal_o2_replication.json`;
- `artifacts/gimbal_o2_uncertainty_calibration.json`;
- `artifacts/gimbal_recovery_robustness_protocol.json`; and
- `artifacts/gimbal_edge_recovery_protocol.json`.

The implementation currently passes 73 automated tests. Generated SVG figures are committed with this report, while the larger experiment artifacts remain local reproducibility products.

11. Conclusion

The project has moved beyond a proportional heuristic and beyond a single model run. It now contains a compact recurrent estimator whose tracking and visibility gains over a model-based analytical controller replicate across rate and position modes and across three training initializations. The benchmark also identifies the boundary of that success: unreachable geometry, command smoothness, off-screen observability, and safe recovery.

The correct present verdict is neither “deployment ready” nor “just a heuristic.” It is a validated synthetic research prototype with a demonstrated temporal inference advantage and an explicit list of unresolved engineering and scientific gates. The next decisive evidence must come from declared mission requirements, recorded platform traces, identified hardware, and physical testing.

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